ZZ-M-1538B
July 13, 1989
SUPERSEDING
ZZ-M-1538A
April 29, 1975 and
Int. Fed. Spec.
ZZ-M-001538(GSA-FSS)
May 21, 1969

FEDERAL SPECIFICATION
MAT, FLOOR: PLASTIC (FOR CHAIRS)

This Federal specification is approved for use by the Commissioner, Federal Supply Service, General Services Administration, for use by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers transparent plastic floor mats for use under chairs.

1.2 Classification. The floor mats shall be of the following types, classes, sizes, and styles. Tolerances are specified in paragraph 3.3.1 and 3.3.2. All dimensions are overall and in inches. Ordering data is specified in paragraph 6.2.

Type I - Plastic sheet, polyvinylchloride with cleats/studs (see figure 1).

Class 1 - With single lip.

Size 1 - 48L x 36W
Thickness a - 0.150 inch

Size 2 - 53L x 45W
Thickness a - 0.150 inch
Thickness b - 0.220 inch

Class 2 - Without lip (rectangular).

Size 3 - 60L x 60W
Thickness a - 0.150 inch

Type II - Plastic sheet, polycarbonate (see figure 1).

Class 1 - With single lip.

Size 2 - 53L - 45W
Thickness c - 0.080 inch

DISTRIBUTION STATEMENT A.
Approved for public release; distribution is unlimited.

FSC 7220

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of this issue in effect on date of this invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Military Specification:

MIL-P-46144 - Plastic Sheet, Polycarbonate

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing and Materials (ASTM) Standard Specification:

ASTM D 2240 - Test Method for Rubber Property - Durometer Hardness

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

3. REQUIREMENTS

3.1 Material.

3.1.1 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580, as amended, to the maximum extent practicable.

3.1.2 Polyvinylchloride (PVC). The type I floor mat shall be made from 100 percent pure PVC. Scrap PVC from inhouse production is acceptable for recycling. Fillers (e.g. calcium carbonate, clay, etc.) are not acceptable.

3.1.2.1 Hardness. The PVC mat material shall have a shore D hardness of 66 $\pm$ 5 when tested in accordance with ASTM D 2240. The readings shall be taken after the presser foot has been in contact with the specimen for 5 seconds. The test specimen shall stablize for a minimum of 24 hours at $70^{\circ} \pm 2^{\circ}$ F prior to conducting the hardness test (see para. 4.4).

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3.1.2.2 Serviceability. The PVC mat material shall not break, crack, curl, show visible permanent indentation or other evidence of damage when tested as specified in 4.5. Service test (4.5) is required at first article testing. A certification of compliance shall be provided with each lot. In addition, a new service test is required on any new PVC supplier's material not previously tested during the contract period. This test must be performed before the new supplier's PVC can be used in production.

3.1.3 Polycarbonate. The type II floor mat shall conform to Type I, class 2 of MIL-P-46144.

3.2 Construction. The mat shall be made of one piece of plastic material specified in 3.1. Lip on class 1 mat shall be centrally located at one end of the mat (across the width direction). All corners of the mat shall be rounded. When laid on a flat surface the mat shall lay flat and shall not show signs of warp or curl.

3.2.1 Type I. The face surface of the mat shall have a uniform textured pattern embossed into the plastic. The underside shall contain a minimum of 144 molded cleats/studs per square foot. The cleats shall run in a straight line, grid pattern (see figure 1) and vary no more than 0.25 inch from being parallel to the edges.

3.2.2 Type II. The top face surface shall have a uniform textured pattern impressed or embossed into the plastic mat. The mat shall have a transparent nonskid coating uniformly applied on the bottom face surface.

3.3 Dimensions and thicknesses. Overall dimensions and thicknesses are as specified in paragraph 1.2. Lip size, and length and width directions are as specified in figure 1.

3.3.1 Thickness tolerances.

Type I
Thickness a - ± 0.008 inch
Thickness b - ± 0.011 inch

Type II
Thickness c - ± 0.010 inch

3.3.2 Length and width tolerances. A tolerance of ± 0.50 inch shall be allowed on overall dimensions.

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3.4 Edge finishing. The edges of the mat shall be straight and evenly cut without burrs.

3.5 Color. The PVC and polycarbonate mat material shall be a colorless, transparent plastic (a slight blue tint is acceptable).

3.6 Workmanship. The mat shall be free from cracks, embedded foreign matter, scratches, bubbles or other defects which affect its appearance or which may impair its serviceability. The finished mat shall conform to the quality and grade of the product established by this specification. Edges shall be straight and uniformly cut, not wavy. For the type I mat, rows of cleats shall be parallel to edges (see 3.2.1.). Cleats shall be evenly formed and securely attached.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article approval. The requirements for first article approval and the responsibility (Government and contractor) for first article testing shall be as specified in the contract. The sample for first article approval test shall consist of one of each item awarded. Each item shall be examined and tested as specified herein for all the requirements of this specification. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production. Approved First Article samples shall serve as a manufacturing standard during the contract and shall be maintained until the last order is shipped, delivered and accepted. New first article samples are required at the start of a new contract.

4.3 Inspection for acceptance. Inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Inspection of materials. In accordance with 4.1 above, the supplier is responsible for insuring that materials are manufactured, tested and inspected to ensure conformance with the requirements of this specification.

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4.3.2 End item inspection. The inspection lot shall be the number of mats of one type, class, size, and style offered for inspection at one time. The sample unit shall be one complete mat.

4.3.2.1 Visual examination. The mat shall be examined for defects listed in table I. The inspection level shall be level I of MIL-STD-105. The acceptable quality level (AQL) shall be 2.5 for major defects and 4.0 for total defects, expressed as defects per hundred units.

TABLE I. Classification of defects.

Defects	Classification Major	Minor
Not type and class specified.	X	
Type I:		
Top face surface not textured uniformly.	X	
Number of cleats/studs not as specified.		X
An average of 1-2 cleats/studs per square foot (on finished mat) missing, broken or not fully formed (do not count cleats split along edge of mat in this calculation).		X
An average of over 2 cleats/studs per square foot (on finished mat) missing, broken, or not fully formed (do not count cleats split along edge of mat in this calculation).	X	
Rows of cleats not parallel to the mat edge.	X	
Type II:		
Top face surface not textured uniformly.	X	
Bottom surface not nonskid coated.	X	
Nonskid coating not uniformly applied.		X
Type I and II:		
Edges noticeably warped or curled.	X	
Material contains cracks, embedded foreign matter scratches, bubbles.	X	
Color and transparency not as specified.	X	
Edges not straight and evenly cut.	X	
Edges are burred.	X	
Corners not completely rounded.	X	
Any discoloration over an area less than 1 sq. inch.		X
Any discoloration over an area greater than 1 sq. inch.	X	

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4.3.2.2 Dimensional examination. The mat shall be examined for dimensional defects. Any measurement deviating from the specified dimension by more than the tolerance shall be considered a defect. The AQL for this examination shall be 2.5 defects per hundred units. The sample size shall be selected in accordance with inspection level S-3 of MIL-STD-105.

4.3.3 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4 percent defective.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect, illegible, improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of the container.
Contents (exterior and interior container)	Number per container more or less than required. Net weight not as specified.

4.4 Hardness test for PVC mat. Testing of the type I PVC mat for hardness shall be conducted as specified in 3.1.2.1. The sample unit shall be one mat. The lot size shall be expressed in units of one mat. The lot shall be unacceptable if one or more sample units fail to meet the test requirement specified. The sample size (number of sample units) shall be in accordance with Table III.

TABLE III. Sample for testing

Lot size	Sample size
800 or less	3
801 thru 22,000	5
22,001 and up	8

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4.5 Service test for PVC mat. Sample mat shall be tested at room temperature (70° ± 2° F). A flat hard surface (wood, tile or concrete) covered with medium pile carpet (carpet cushion not mandatory) shall be used for this test. The wheels shall be 2-1/2 inch Star Flite* metal ball casters. The chair (with a four prong base) shall have an applied force of 210 ± 10 lbs. The base of the chair shall rotate in a circular motion 10 to 15 revolutions per minute, rotating clockwise for approximately one minute, then pausing for 15 to 20 seconds, then rotating counter-clockwise for approximately one minute. This alternating procedure shall continue for a minimum of 45 hours in as nearly the same place on the mat as possible. The mat shall be examined for conformance with 3.1.2.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. Mats are intended for use under chairs with casters, over carpeted floors, to increase chair mobility and decrease carpet wear. Type I, PVC, thickness a (0.150 inch) mats and type II polycarbonate mats are designed for use over low to medium pile carpet. Type I, PVC, thickness b (0.220 inch) mats are designed for use over medium to high pile carpets. Chairs with metal casters or small diameter, hard plastic wheels should not be used; the concentration of weight over a small area of the mat (pinpoint loading), i.e., desk leg, should be avoided. (Although metal casters are not recommended for normal use on mats described by this specification, they are required in paragraph 4.5 in order for the Government to obtain accelerated service testing information).

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, size and date of this specification
(b) Type, class, size, and thickness required (see 1.2).
(c) Packaging, packing and marking required (see 5.1)

* Star Flite casters (Model O-25CH) may be obtained from Master Casters, Inc., 9200 Inman Avenue, Cleveland, Ohio 44105, phone 1-800-323-5513.

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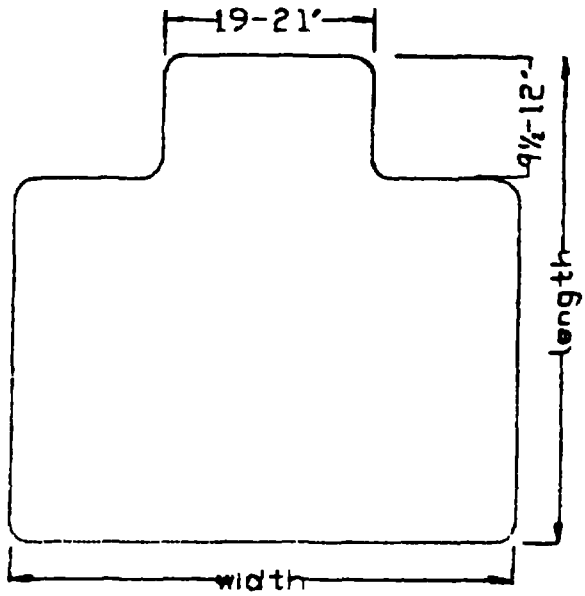
6.3 Reference part numbering system. This system is to be used as guide only. Not all combinations of types, classes, sizes and thicknesses are available. See 1.2 Classification.

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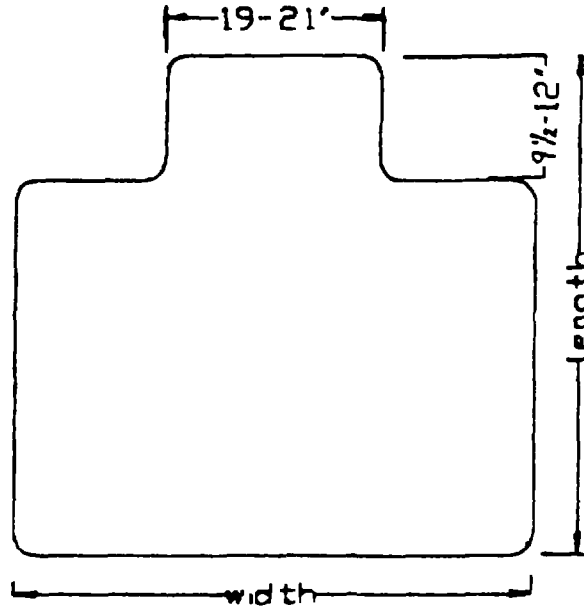
Thickness:	A	Thickness a 0.150
	B	b 0.220
	C	c 0.080
Size.	1	Size 1 48 X 36
	2	2 53 X 45
	3	3 60 X 60
Class:	1	Class 1 single ply
	2	2 w/o ply
Type:	1	Type I PVC mat
	2	II Polycarbonate mat

Preparing activity
GSA FSS

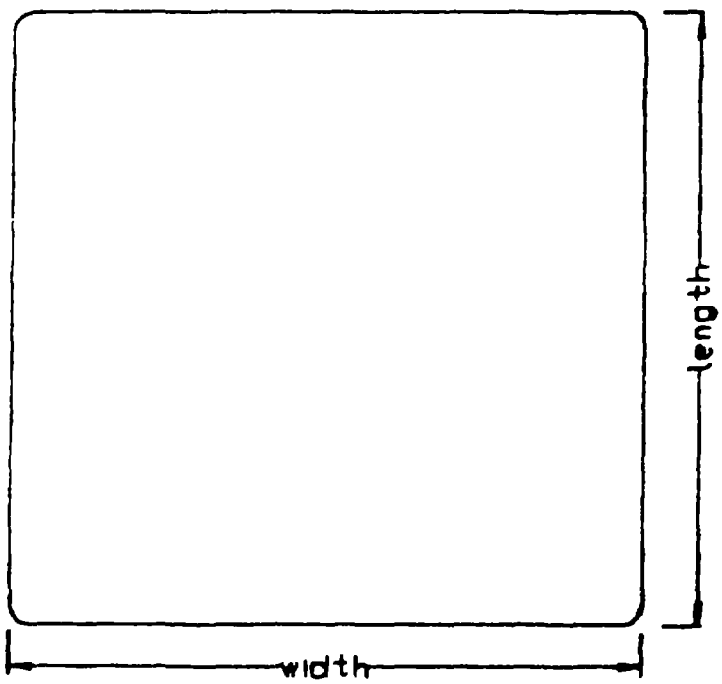
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Type I, Class 1



Type II, Class 1



Type I, Class 2

NOTES:
1 Dots show required pattern for cleats/studs. They are not to scale (see para. 3.2.1).

2 If a different lip size is required, it shall be as specified in contract (see 6.2).

FIGURE 1
Pattern outlines and cleat/stud arrangement for chair mats.

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